

Depth First Search

0 -> 1, 2

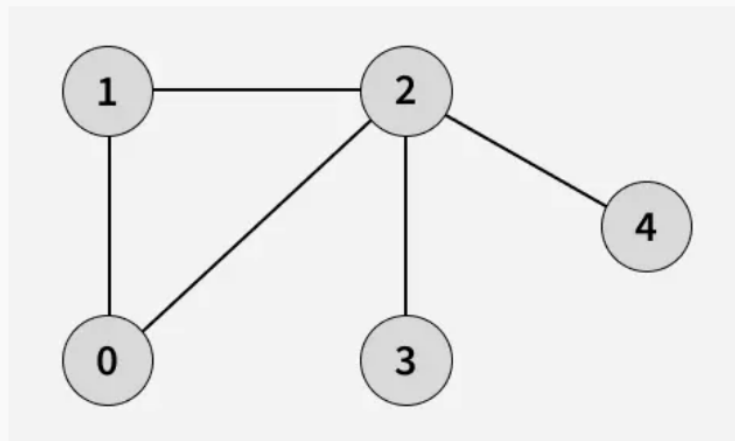
1 -> 0, 2

2 -> 0, 1, 3, 4

3 -> 2

4 -> 2

Input: $adj[][] = [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]]$



Output: [0, 1, 2, 3, 4]

Explanation: The source vertex is 0. We visit it first, then we visit its adjacent.

Start at 0: Mark as visited. Print 0.

Move to 1: Mark as visited. Print 1.

Move to 2: Mark as visited. Print 2.

Move to 3: Mark as visited. Print 3.(backtrack to 2)

Move to 4: Mark as visited. Print 4(backtrack to 2, then backtrack to 1, then to 0).

Input: $adj[][] = [[1, 2], [0, 2], [0, 1, 3, 4], [2], [2]]$

Output: [0, 1, 2, 3, 4]